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Local Lattice Dynamics in the $Mg_{0.5}Al_{0.5}B_2$ Superconductor

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Abstract We have studied temperature evolution of the local as well as the average crystal structure of MgB_2 and $Mg_{0.5}Al_{0.5}B_2$ using real-space atomic pair distribution function (PDF) measured by high resolution neutron powder diffraction in a wide temperature range of $T = 10$–600 K. The mean square relative displacements (MSRD) of atomic B–B, B–Mg (B–Al) pairs are compared with mean-square displacements (MSD) to calculate atomic correlations. In spite of the enhanced atomic disorder in $Mg_{0.5}Al_{0.5}B_2$, where the boron–boron, and boron–magnesium pair motions are found to be small, we find that the same atomic correlations in MgB_2 assume even slightly lower values and remain nearly constant in a wide temperature range of 0–600 K. This anomalous behavior and its physical interpretation provoke new questions on our understanding to the local lattice dynamics in this material.

Keywords MgB_2 · Neutron diffraction

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1 Introduction

MgB_2 is the simplest system to investigate the quantum mechanism of the formation of a superconducting condensate with a critical temperature of $T_c \sim 40$ K [1], a factor two higher than in all other known intermetallic superconductors. There is now a general consensus about MgB_2 being a high T_c multiband superconductor where the electron-phonon interaction constructs a weak pairing in the π channel and a strong pairing in the σ-channel [2–13]. This strong electron-phonon coupling, driven by the interaction between electronic carriers in the 2D σ band with boron $p_{x;y}$ character and the zone center E_{2g} phonon mode [12–17], is reflected in a Kohn anomaly in the phonon dispersion related to the size of the small 2D tubular Fermi surfaces [18]. The proximity of the Fermi level to the Van Hove singularity (VHs) and to the band edge discloses a new scenario where the large amplitude of the expected boron zero point lattice fluctuations [7, 12, 15, 19–21] induces large fluctuations of the same order of the separation between the VHs, the gap edge and the Fermi level itself. Although the amplitude of the lattice fluctuations seems thus to be highly relevant for the superconductivity in MgB_2, there is a lack of experimental information on this key point. Furthermore, although the average structure (P6/mmm) of the MgB_2 system has been exhaustively investigated, there is not yet any study on the local structure, since the usual X-ray local probes, as EXAFS, is limited by poor scattering amplitude due to light atoms.

In view of this, we have investigated the local as well as the average structure of MgB_2 and $Mg_{0.5}Al_{0.5}B_2$ using high resolution neutron diffraction to obtain the pair distribution function (PDF). We focused here on the atomic pairs correlations founded by comparing the mean-square displacements (MSD) and the mean-square relative displace-

ments (MSRD) of the boron–boron (ρ_{B-B}), and boron–magnesium (aluminium) ($\rho_{B-Mg/Al}$) pair motions. Comparing the atomic correlations in MgB_2 and $Mg_{0.5}Al_{0.5}B_2$, we have been able to investigate the effects on the local structure of the two main microscopic mechanisms for the T_c decrease produced by the Al substitution [4]: (i) the filling of the electronic states in the 2D σ band of $p_{x;y}$-orbitals within the boron layers and the resulting decrease in the density of states at the Fermi level, and (ii) the increased disorder of the inter-band scattering induced by lattice disorder on the atomic scale. We find that in both MgB_2 and $Mg_{0.5}Al_{0.5}B_2$, the atomic correlations for B–B (ρ_{B-B}) and B–Mg/Al ($\rho_{B-Mg/Al}$) bonds show small ($\sim$0.2) positive and anomalous nearly temperature independent behavior.

2 Experimental

Polycrystalline samples of MgB_2 and $Mg_{0.5}Al_{0.5}B_2$ were synthesized at high temperature by direct reaction of the elements in a tantalum crucible under argon atmosphere using pure ^{11}B isotopes. Time-of-flight neutron powder diffraction data were collected on the NPDF diffractometer at the Manuel Lujan, Jr., Neutron Scattering Center (LANSCE) at Los Alamos National Laboratory. The NPDF diffractometer, with its high neutron flux and backscattering detector modules provides a high resolution [22], permitting access to a wide range of momentum transfer with sufficient counting statistics making it ideal for PDF studies. The powdered samples were sealed inside extruded cylindrical vanadium containers. These were mounted on the stage of a cryo-furnace with and without heat-shield for $T < 300$ K and $T > 300$ K, respectively. The scattering data from the empty cryo-furnace, with and without heat-shield, an empty container mounted on the cryo-furnace, and the empty instrument were also collected, allowing us to assess and subtract instrumental backgrounds. The scattering from a vanadium rod was also measured to allow the data to be normalized for the incident spectrum and detector efficiencies. We collected each diffraction spectrum up to the high momentum transfer of $Q = 40$ Å^{-1} in 3 hours. The high resolution diffraction spectrum so obtained presents both the Bragg peaks and the diffuse scattering. While the Bragg peaks reflect the long-range order of the crystalline samples, the oscillating diffuse scattering contains local structural information including the correlated dynamics [23, 24]. The PDF is obtained from a Fourier transform of the powder diffraction spectrum (Bragg peaks + diffuse scattering) [25]. It consists of a series of peaks, the positions of which give the distances of atom pairs in the real space, while the ideal width of these peaks is due both to relative thermal atomic motion and to static disorder.

This permits the study of the effects of lattice fluctuations on PDF peak widths and yields information on both mean-square displacements and the mean-square relative displacements of atom pairs and their correlations. Standard data corrections [25] were carried out using the program PDFGETN [26]. After being corrected, the data were normalized by the total scattering cross section of the sample to yield the total scattering structure function $S(Q)$. Afterwards, the total scattering structure function $S(Q)$ is converted to the PDF, $G(r)$, by means of a sine Fourier transform according to the relation:

$$G(r) = 4\pi r(\rho(r) - \rho_0) = \frac{2}{\pi}\int_0^\infty Q[S(Q) - 1]\sin(Qr)\,dQ. \quad (1)$$

We modeled the PDF using a structural model that takes advantage of the definition of the radial distribution function RDF $R(r)$, namely:

$$\mathrm{R}(r) = \sum_i \sum_{i \neq j} \frac{b_i b_j}{\langle b^2 \rangle} \delta(r - r_{ij}), \quad (2)$$

where b_i is the scattering length of the ith atom, $\langle b \rangle$ is the scattering length averaged over the sample composition, $r_{ij} = |\mathbf{r}_i - \mathbf{r}_j|$ is the distance between the ith and the jth atoms, and the sums are taken over all the atoms in the sample. Before being compared to the data, the calculated $G(r)$ is convoluted with a termination function, $\sin(Q_{\max}r)/r$ to account for the effects of the finite data collection range. Fundamental lattice information, such as the average crystal structure, the lattice constants, a scale factor, and the refined atomic (thermal) displacement parameters, can now be extracted from the PDF by using the PDF refinement program PDFFIT [27] that is based on a least-squares approach to fit the PDF profile. The mean square displacement of atom i along the major axes x, y and z, $\sigma^2(i_x)$, $\sigma^2(i_y)$, $\sigma^2(i_z)$, are defined as

$$\sigma^2(i_\alpha) = \left\langle [\mathbf{u}_i \cdot \hat{\mathbf{r}}_\alpha]^2 \right\rangle, \quad (3)$$

where $\mathbf{u}_i$ is the lattice displacement of atom i from its average position 28 and $\hat{\mathbf{r}}_\alpha$ is the unit vector pointing along the direction $\alpha = x, y, z$. Due to the geometry of these compounds, the two boron atoms for the unit cell are equivalent and $\sigma^2(i_x) = \sigma^2(i_y)$ for all the atoms, so that only four parameters were needed, namely, $\sigma^2(B_{xy})$, $\sigma^2(B_z)$, $\sigma^2(Mg_{xy})$, and $\sigma^2(Mg_z)$. We found the same mean square displacements for Mg and Al atom in $Mg_{0.5}Al_{0.5}B_2$.

3 Results

In Fig. 1, we have shown the reduced scattering structure function $Q[S(Q) - 1]$ for the MgB_2 and $Mg_{0.5}Al_{0.5}B_2$ at

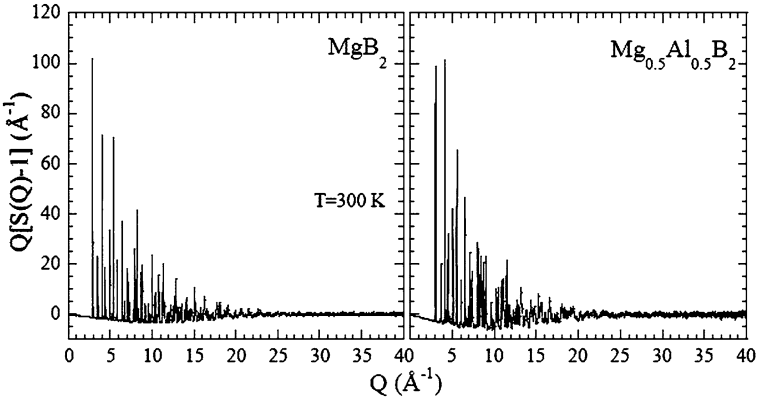


Fig. 1 The reduced structure function $Q[S(Q)-1]$ for MgB_2 (*left panel*) and $Mg_{0.5}Al_{0.5}B_2$ (*right panel*) measured at 300 K

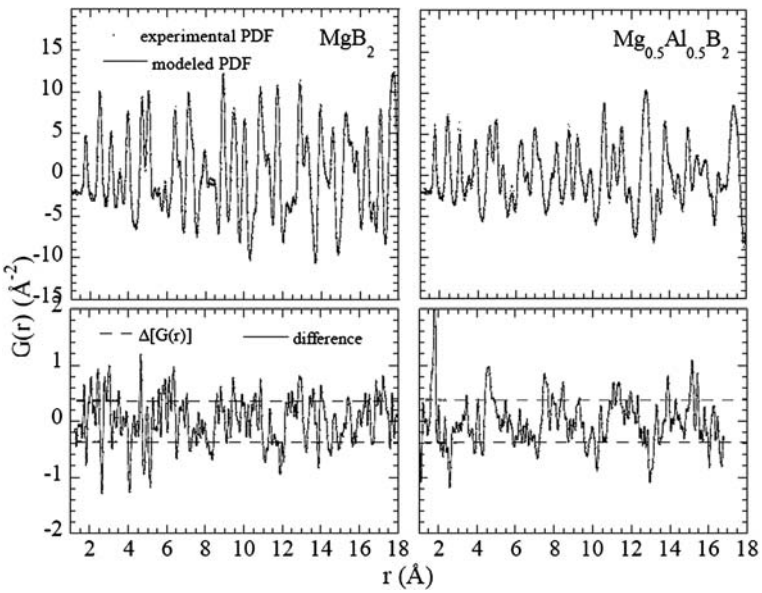


Fig. 2 (*Upper panels*) The PDF, $G(r)$, obtained from (1) for MgB_2 (*left panel*) and $Mg_{0.5}Al_{0.5}B_2$ (*right panel*) measured at 300 K (*dots*), with the structure refinement curves obtained by a least-squares approach (*solid lines*). (*Lower panels*) The difference curves (*solid lines*) of the experimental PDFs with the modeled fits and the standard deviations on the data $\Delta[G(r)]$ (*dashed lines*) are shown. We can observe that most of the fluctuations in the difference curves are within $\Delta[G(r)]$

$T = 300$ K, while the corresponding reduced PDFs, $G(r)$, obtained using (1), are shown in the upper panels of Fig. 2.

The features of the NPDF diffractometer allowed us to obtain high quality PDFs as can be noted by inspecting the modeled fit (solid line) of the $G(r)$ in the upper panels (Fig. 2). This can be seen also in the lower panels of Fig. 2 where most of the fluctuations in the difference curves (solid lines) are within the standard deviation of the data $\Delta[G(r)]$ (dashed lines above and below the difference curves). The PDFs have been fit over the range 1–18 Å using a hexagonal crystal structure (space group P6/mmm), obtaining very good fits at all temperatures. The Bragg peaks are clearly persistent up to 25 $Å^{-1}$, reflecting both the long-range order of the crystalline samples and the small amount of positional (vibrational or static) disorder of the atoms around their average positions. The lattice parameters at $T = 300$ K for MgB_2 ($T_c \sim 39$ K) and $Mg_{0.5}Al_{0.5}B_2$ were found to be $a = 3.08505(4)$ Å, $c = 3.5218(1)$ Å and $a = 3.05457(5)$ Å, $c = 3.37863(10)$ Å, respectively. The mean square displacements $\sigma^2(B_{xy})$, $\sigma^2(B_z)$, $\sigma^2(Mg_{xy}/Al_{xy})$, and $\sigma^2(Mg_z/Al_z)$ for MgB_2 (open circles) and $Mg_{0.5}Al_{0.5}B_2$ (open squares) are shown in Fig. 3.

In order to extract information about the phonon spectrum, we fit each MSD with a simple Einstein model

$$\sigma^2(i_\alpha) = \frac{\hbar}{m_i \omega_{i\alpha}} \left[\frac{1}{2} + n[\omega(i_\alpha)] \right] + \sigma_0^2(i_\alpha), \tag{4}$$

where $\sigma_0^2(i_\alpha)$ takes into account the static disorder, M_i is atomic mass of the atom i and $\omega(i_\alpha)$ provided an estimate of the vibrational frequency of the atom i along the direction α. The quantity $n[x]$ is the Bose thermal factor $n[x] = [\exp(x/K_B T) - 1]^{-1}$. The good fits indicate that a simple Einstein model seems to describe quite well the lattice vibrations in these compounds. Moreover, the Einstein fits were obtained by simply using nominal atomic masses $M_{Mg} = 24.3$ a.m.u. for $\sigma^2(Mg)$, $M_B = 10.81$ a.m.u. for $\sigma^2(B)$ and $M_{Al} = 26.98$ a.m.u. for $\sigma^2(Al)$. This means that we are implicitly assuming the magnesium and boron vibrations being decoupled. A more quantitative check of this assumption, obtained calculating the phonon dispersion of MgB_2 within

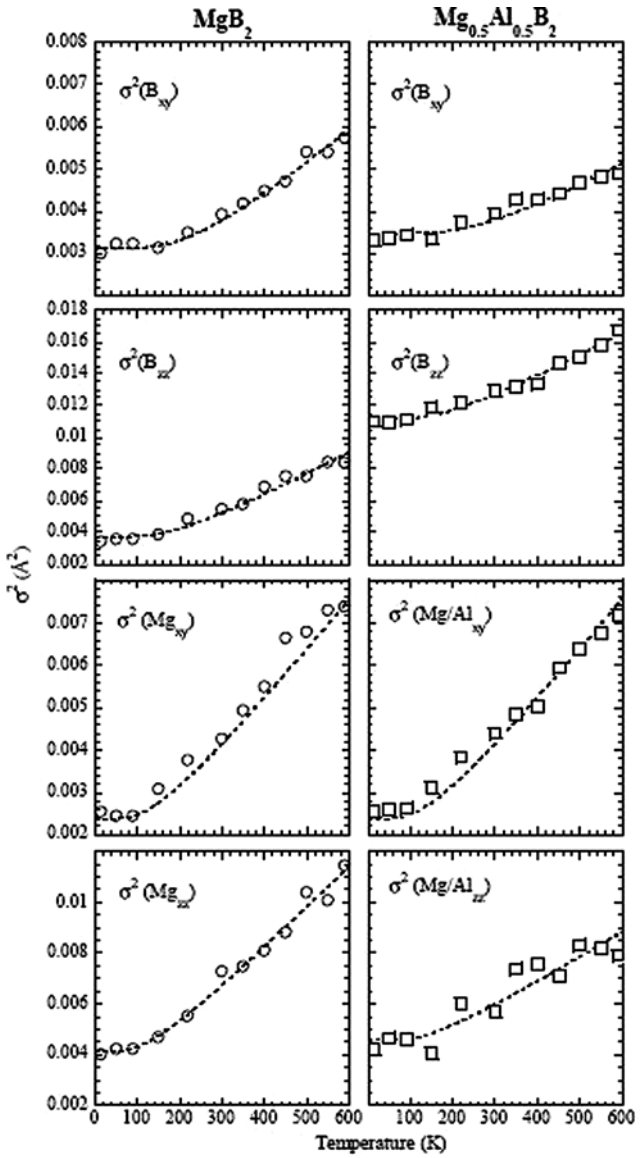


Fig. 3 Anisotropic in plane $\sigma^2(B_{xy})$, $\sigma^2(Mg_{xy}/Al_{xy})$ and along the c-axis $\sigma^2(B_z)$, $\sigma^2(Mg_z/Al_z)$ mean-square displacements for the B and Mg(Al) atoms, as a function of the temperature in MgB_2 (*circles* in *left column* panels) and $Mg_{0.5}Al_{0.5}B_2$ (*squares* in *right column* panel). The *open circles* and *squares* represent the PDFFIT refined values, while the *dotted lines* represent the modelled data in the Einstein model. The *error bars* in the mean-square displacements are smaller than the size of the used symbols

a force constant shell model, can be found in [29]. The values of the fitting parameters, $\omega(i_\alpha)$ and $\sigma_0^2(i_\alpha)$, are reported in Table 1, and the σ^2 vs. T fitting curves are represented by dotted lines in the Fig. 3. The different values of $\omega(i_\alpha)$ represent the different energy range of the phonon spectra associated with the boron and magnesium (aluminium) in-plane and out-of-plane lattice vibrations, and they are in good agreement with the corresponding spectra reported in [30]. Upon inspection of the σ_0 values, one will note the larger static disorder term σ_0 for the B and Mg(Al) atoms motion

Table 1 Effective phonon frequencies $\omega(i_\alpha)$ and static disorder contributions $\sigma_0^2(i_\alpha)$ for the in-plane and out-of-plane B and Mg/Al displacements as obtained by the Einstein model fit (4) of $\sigma^2(i_\alpha)$

	MgB_2	$Mg_{0.5}Al_{0.5}B_2$
$\omega(B_{xy})$ (K)	703 ± 10	876 ± 46
$\omega(B_{zz})$ (K)	572 ± 15	560 ± 23
$\omega(Mg/Al_{xy})$ (K)	398 ± 6	400 ± 13
$\omega(Mg/Al_{zz})$ (K)	350 ± 9	415 ± 31
$\sigma_0^2(B_{xy})$ (Å^2)	$(5.11 \pm 5.11) \times 10^{-5}$	0.0011 ± 0.0002
$\sigma_0^2(B_{zz})$ (Å^2)	$(9.10 \pm 9.10) \times 10^{-5}$	0.0074 ± 0.0004
$\sigma_0^2(Mg/Al_{xy})$ (Å^2)	$(7.35 \pm 7.35) \times 10^{-5}$	0.0002 ± 0.0002
$\sigma_0^2(Mg/Al_{zz})$ (Å^2)	0.0013 ± 0.0003	0.0024 ± 0.0004

in $Mg_{0.5}Al_{0.5}B_2$ than in MgB_2 where we find contribution of the disorder only for the magnesium out-of-plane lattice vibration ($\sigma^2(Mg_z) \sim 0.0013$ Å^2) while it is found to be negligible for the other modes.

The PDF technique is sensitive to the correlations in the atomic dynamics that contain some additional details about the underlying interatomic potentials [23, 24, 28, 31, 32]. Here, we explore the motional correlations in the two compounds using the PDF data. The Gaussian width σ_{ij} of the PDF peaks is directly related to the mean-square relative displacement (MSRD) of atomic pairs projected onto the vector joining the atom pairs 28, that is

$$\sigma_{ij}^2 = \left\langle [(\mathbf{u}_i - \mathbf{u}_j) \cdot \hat{\mathbf{r}}_{ij}]^2 \right\rangle, \tag{5}$$

where $\mathbf{u}_i$ and $\mathbf{u}_j$ are the lattice displacements of atoms i and j from their average positions, $\hat{\mathbf{r}}_{ij}$ is the unit vector connecting atoms i and j, and the angular brackets indicate an ensemble average [28]. As a general consideration, we would like to stress once more that while $\sigma_0^2(i_\alpha)$ probes the absolute magnitude of the single atom mean-square displacement, σ_{ij}^2 provides information about the *correlation* between the lattice displacements of atom pairs. Let us consider for instance the case of $\sigma_{\mathrm{B-B}}^2$ which involves only boron in-plane lattice fluctuations. We can identify three limiting behaviors for this quantity: (i) perfectly in-phase lattice motion; (ii) perfectly opposite-phase motion; and (iii) completely independent motion. In the first case, it is easy to see that $\sigma_{\mathrm{B-B}}^2 = 0$, while $\sigma_{\mathrm{B-B}}^2 = 2\sigma^2(B_{xy})$ when the nearest neighbor boron lattice displacements are uncorrelated, and $\sigma_{\mathrm{B-B}}^2 = 4\sigma^2(B_{xy})$ when they have opposite phase. To formalize this, we rearrange (5) as

$$\sigma_{ij}^2 = \left\langle [\mathbf{u}_i \cdot \hat{\mathbf{r}}_{ij}]^2 \right\rangle + \left\langle [\mathbf{u}_j \cdot \hat{\mathbf{r}}_{ij}]^2 \right\rangle - 2\langle (\mathbf{u}_i \cdot \hat{\mathbf{r}}_{ij})(\mathbf{u}_j \cdot \hat{\mathbf{r}}_{ij}) \rangle. \tag{6}$$

Here, the first two terms are related to mean-square thermal displacement of atoms i and j projected along $\hat{\mathbf{r}}_{ij}$, while the third term is a displacement correlation function, which carries information about the motional correlations. It is now

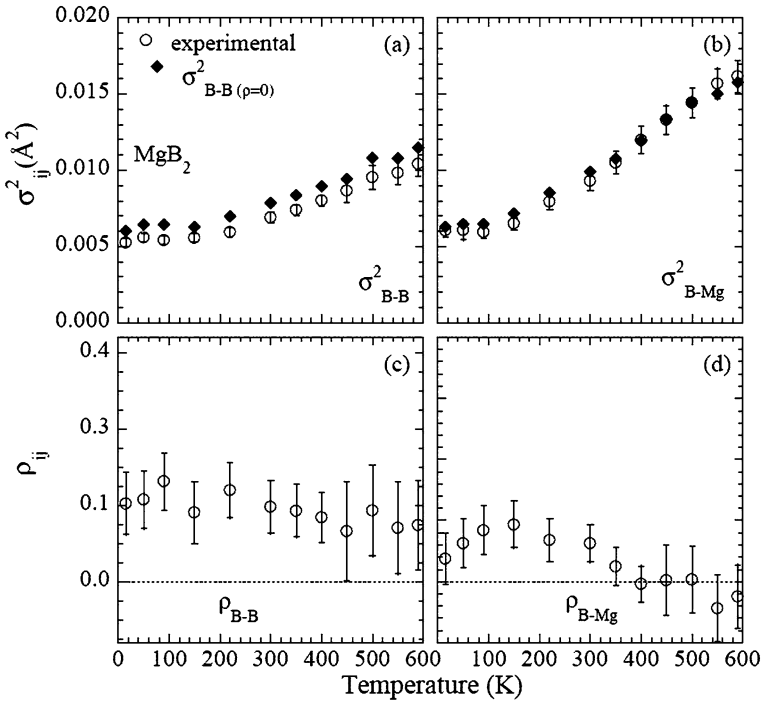


Fig. 4 Mean-square relative lattice displacements σ^2_{B-B} (**a**), σ^2_{B-Mg} (**b**), as extracted from the width of the PDF peaks and the corresponding correlation function ρ_{B-B} (**c**), ρ_{B-Mg} (**d**) (*open circles*) for MgB_2. The *filled diamonds* represent lattice displacements σ^2_{B-B}, σ^2_{B-Mg} for completely uncorrelated lattice motion, i.e., $\rho_{ij} = 0$

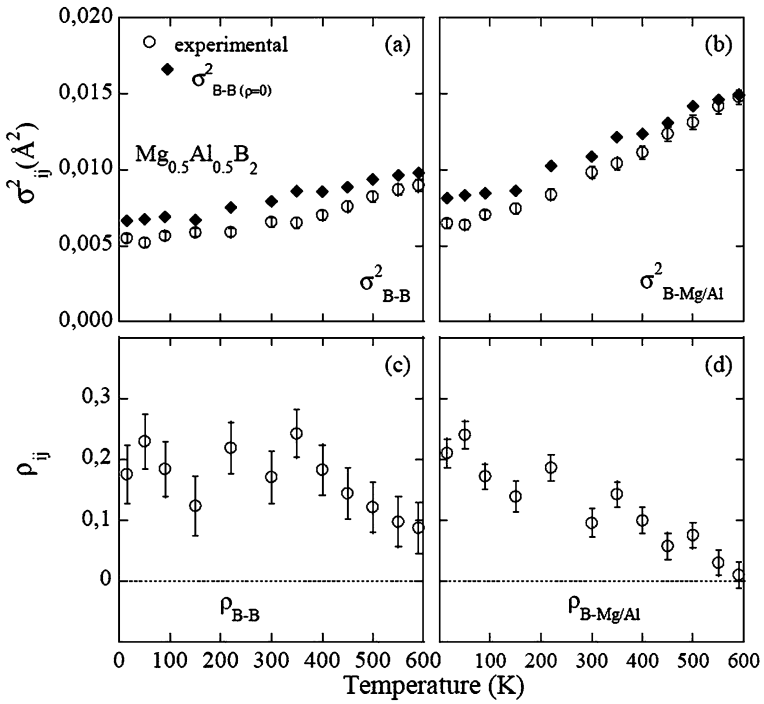


Fig. 5 The same quantities of Fig. 4, namely, mean-square relative lattice displacements σ^2_{B-B} (**a**), $\sigma^2_{B-Mg/Al}$ (**b**), with the corresponding correlation function ρ_{B-B} (**c**), $\rho_{B-Mg/Al}$ (**d**) (*open circles*) and lattice displacements σ^2_{B-B}, $\sigma^2_{B-Mg/Al}$ for completely uncorrelated lattice motion, i.e., $\rho_{ij} = 0$ are here illustrated for the $Mg_{0.5}Al_{0.5}B_2$

useful to quantify the degree of correlation by introducing the dimensionless correlation parameter ρ_{ij} defined as [23, 24, 31, 33]:

$$\sigma^2_{ij} = \sigma^2(i_j) + \sigma^2(j_i) - 2\sigma(i_j)\sigma(j_i)\rho_{ij}, \tag{7}$$

where $\sigma^2(i_j) = \langle[\mathbf{u}_i \cdot \hat{\mathbf{r}}_{ij}]^2\rangle$. Positive values of ρ describe a situation where the atoms move in phase, so that the resulting value of σ^2_{ij} is smaller than for the uncorrelated case. On the other hand, a predominance of opposite phase atomic vibrations should result in $\rho < 0$ and in a PDF peak width σ^2_{ij} larger than the uncorrelated case. It is important to note that the correlation function ρ_{ij} in (7) expresses the degree of correlation between the *total* atomic displacements. Using (7), the correlation parameter can be calculated from the total width of the PDF peak as:

$$\rho_{ij} = \frac{(\sigma^2(i_j) + \sigma^2(j_i)) - \sigma_{ij}^2}{2\sigma(i_j)\sigma(j_i)}. \quad (8)$$

In our analysis, we focus on the width of the nearest neighbor boron-boron PDF peak, $\sigma^2_{\mathrm{B-B}}$, and on the nearest neighbor magnesium (aluminium)-boron peak, $\sigma^2_{\mathrm{B-Mg/Al}}$. These are well resolved single-component peaks in the PDF whose width directly yields correlated dynamical information [24]. The Gaussian widths $\sigma^2_{\mathrm{B-B}}$, $\sigma^2_{\mathrm{B-Mg/Al}}$, as measured by the PDF data are shown in the upper panels of Figs. 4 and 5 (open circles) for MgB_2 and $Mg_{0.5}Al_{0.5}B_2$. The correlation factor ρ_{ij} as extracted from the PDF data $\sigma^2_{\mathrm{B-B}}, \sigma^2_{\mathrm{B-Mg/Al}}$, and from the single atom mean-square lattice displacements $\sigma^2(i_\alpha)$ is shown in the lower panels of Figs. 4 (MgB_2) and 5 ($Mg_{0.5}Al_{0.5}B_2$). We find a positive correlation factor for both $\rho_{\mathrm{B-B}} \sim 0$, 1, and $\rho_{\mathrm{B-B}} \sim 0$, 1, indicating a slight predominance of the in-phase B–B and B–Mg/Al lattice displacements in this experimental probe. Positive values of ρ_{ij} are commonly reported in a variety of materials. The intuitive explanation is that the in-phase phonon modes (acoustic, low optical branch modes) are generally less stiff than the opposite-phase optical ones. Our reported values of $\rho_{\mathrm{B-B}}$; $\rho_{\mathrm{B-Mg/Al}} \sim 0.1$ are, however, much smaller than the correlation factors commonly found in other covalently bonded materials [23, 24, 31, 33].

4 Conclusion

In this work, we have investigated local lattice properties of the MgB_2 and $Mg_{0.5}Al_{0.5}B_2$ compounds paying special attention on the lattice dynamics and the correlations in the B–B and B–Mg/Al first neighbor atomic pair motion. We have used the real space PDF obtained from high resolution neutron diffraction to study the effects of the lattice vibrations on the PDF peak widths. The PDF peaks in well ordered crystals, such as the present case, yield important information about the underlying atomic potentials through the correlated local lattice dynamics. The analysis of the PDF peak linewidths permits to evaluate the correlation for both the nearest neighbor B–B and B–Mg/Al atomic pairs. We find a small positive correlation factor $\rho_{\mathrm{B-B}} \sim 0.1$ and $\rho_{\mathrm{B-B}} \sim 0.1$, nearly temperature independent, indicating a weakly prevalent in-phase relative atomic motion.

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